

Embedding Consistency Filtering (ECF) defense mechanism

Four interconnected components make up the modular defense framework that is the Embedding Consistency Filtering (ECF) approach. By spotting statistical and structural irregularities in client updates, these modules work together to detect and counteract covert untargeted poisoning attacks (like StegaPoison) in federated recommender systems.

1. Temporal Embedding Drift Monitoring

To identify unusual update behavior, the embedding updates for each client are monitored over time.

$$\Delta E = \|E_t - E_{t-1}\|_2$$

To ascertain whether the change falls outside of $[\mu - k\sigma, \mu + k\sigma]$, the Euclidean norm ΔE is compared to a rolling statistical profile of previous updates (mean μ , standard deviation σ) for every round.

where, k is a tunable sensitivity parameter (usually between 1 and 3 in practice). Larger k (e.g., 3): Accepts a wider range of variation $\rightarrow$ less sensitive, fewer false positives, but may miss subtle attacks. Smaller k (e.g., 1 or 1.5): Accepts only small deviations $\rightarrow$ more sensitive, detects more anomalies, but may raise false alarms.

Finding abrupt or significant changes in client behavior that are consistent with untargeted poisoning is the goal.

Reference: Concept drift tracking and temporal anomaly detection in federated systems are the foundations of this approach (Feldhans, et al, 2021).

2. Interdimensional Consistency Check (IDC)

Structured perturbations concealed across embedding dimensions (such as alterations based on PVD) are detected by this module.

Calculate the difference vectors:

$$D_{ij} = |E_i - E_j|$$

for every pair of dimensions (i, j) .

Utilize the following to gauge deviation from reference norms:

- Kullback-Leibler Divergence (KL-D)
- Earth Mover's Distance

The statistical distribution of interdimensional differences is evaluated using these distance measures to see if it departs from the generally positive historical trend.

Finding concealed disturbance patterns introduced by steganographic techniques is the goal.

Reference: Motivated by statistical anomaly identification in adversarial learning and steganalysis metrics (Li et al, 2021).

3. Compatibility Drop Estimator (CDE)

This module measures the *semantic impact* of each client's update by simulating it against the current global model.

- Use cosine similarity to calculate expected compatibility:

$$S = \cos(\theta) = \frac{E_{\text{user}} \cdot E_{\text{item}}}{\|E_{\text{user}}\| \|E_{\text{item}}\|}$$

- Prior to and following the client's update, measure the change in S . Suspicious updates are indicated by large declines in score for normally compatible user-item combinations.

The goal is to identify updates that weaken user-item alignment, which is a feature of mirror-shift attacks.

Reference: Based on online model simulation methods and adversarial compatibility measures (Shams and Leith, 2021).

4. Credibility-Weighted Aggregation

Each client receives soft credibility scores $C_i \in [0, 1]$ from all previous modules, which affect the final aggregation:

$$G_{\text{new}} = \sum C_i \cdot W_i$$

where C_i represents embedding integrity and W_i is the client's weight update. In order to maintain robustness and fairness, this avoids binary rejection and instead lessens the influence of questionable clients.

The goal is to handle malicious clients with grace while preserving valuable updates.

Reference: This idea builds on trust-weighted federated learning and robust aggregation approaches (Liu et al, 2023; Pillutla et al, 2022).